In [37]:
```
if (!require("pacman")) install.packages("pacman")
pacman::p_load(knitr, tidyverse, ggplot2, dplyr, lubridate, stringr, readxl,
source("../data-build/functions.R")
source("../data-build/rating_variables.R")

ma_final <- read_csv("../../hwk3/data/output/final-ma-data.csv")
```

```
Rows: 440877 Columns: 110
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (16): contractid, state, county, org_type, plan_type, partd, snp, eghp,
...
dbl (94): planid, fips, n_nonmiss, avg_enrollment, sd_enrollment, min_enrol
l...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this mes
sage.
```

Question 1

In [2]:
```
summary_table <- ma_final %>%
    filter(!is.na(Star_Rating)) %>%
  mutate(plan_market_share = avg_enrollment / avg_eligibles) %>%
  group_by(year) %>%
  summarize(
    `Mean Star Rating` = mean(Star_Rating, na.rm = TRUE),
    `Mean Enrollment` = mean(avg_enrollment, na.rm = TRUE),
    `Mean Market Share` = mean(plan_market_share, na.rm = TRUE),
    `Plan Count` = n()
  )

summary_table
```

A tibble: 6 × 5

year	Mean Star Rating	Mean Enrollment	Mean Market Share	Plan Count
<dbl>	<dbl>	<dbl>	<dbl>	<int>
2010	2.969968	345.9702	0.01245348	60519
2011	3.255317	388.3079	0.01506869	55249
2012	3.359556	410.6486	0.01656644	58169
2013	3.548875	406.7334	0.01631573	64174
2014	3.806049	451.1140	0.01785064	58636
2015	3.961542	482.5712	0.01930427	60819

Question 2

In [3]:
```
# Create table for unrated plans
summary_no_stars <- ma_final %>%
 filter(is.na(Star_Rating)) %>%
  mutate(plan_market_share = avg_enrollment / avg_eligibles) %>%
  group_by(year) %>%
  summarize(
    `Mean Enrollment` = mean(avg_enrollment, na.rm = TRUE),
    `Mean Market Share` = mean(plan_market_share, na.rm = TRUE),
    `Count of Plans` = n()
  )

summary_no_stars
```

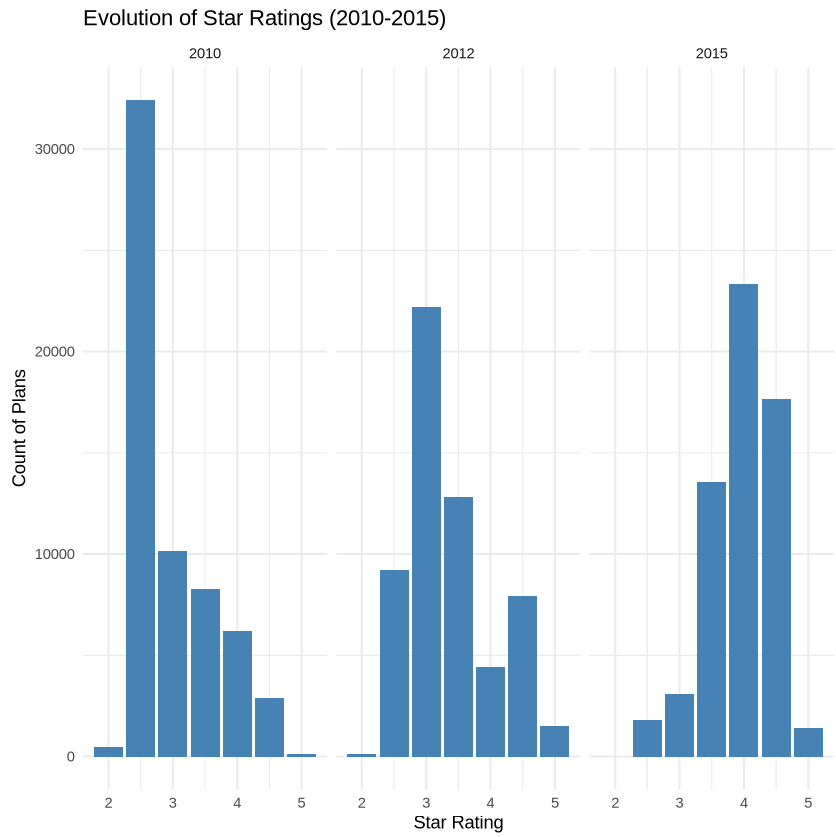
A tibble: 6 × 4

year	Mean Enrollment	Mean Market Share	Count of Plans
<dbl>	<dbl>	<dbl>	<int>
2010	107.2581	0.009183432	49431
2011	167.8986	0.013221663	12754
2012	172.7340	0.011860378	9077
2013	213.5044	0.017416414	3661
2014	257.5963	0.015849555	3708
2015	314.2103	0.008666974	4680

Question 3

In [11]:
```
ma_final %>%
  filter(year %in% c(2010, 2012, 2015)) %>%
  ggplot(aes(x = Star_Rating)) +
  geom_bar(fill = "steelblue") +
  facet_wrap(~year, nrow = 1) +
  labs(
    title = "Evolution of Star Ratings (2010-2015)",
    x = "Star Rating",
    y = "Count of Plans"
  ) +
  theme_minimal()
```

```
Warning message:
“Removed 63188 rows containing non-finite values (`stat_count()`).”
```



Between 2010 and 2015, the distribution of MA star ratings saw a significant rightward shift. In 2010, the market was dominated by lower-rated plans, with 2.5 stars being the most frequent at 30,000+ occurrences. By 2012, the distribution began to center with 3.0 being the most common rating. 2015 shows the distribution had transitioned to a high-quality skew where 4.0 plans were the highest count of plans, while the count of 2.5 star plans dropped to below 5,000.

Question 4

In [38]:
```
reg_data <- ma_final %>%
  group_by(year, contractid, planid) %>%
  summarize(
    star_rating = mean(Star_Rating, na.rm = TRUE),
    plan_enrollment = sum(avg_enrollment, na.rm = TRUE),
    total_market_enrollment = sum(avg_enrolled, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(market_share = plan_enrollment / total_market_enrollment) %>%
  filter(!is.na(star_rating), !is.na(market_share)) %>%
# Create star rating factor (excluding 2.5 or below)
  mutate(star_cat = case_when(
    star_rating <= 2.5 ~ "stars_2.5_or_below",
    star_rating == 3.0 ~ "stars_3.0",
    star_rating == 3.5 ~ "stars_3.5",
    star_rating == 4.0 ~ "stars_4.0",
    star_rating >= 4.5 ~ "stars_4.5_plus"
```

```
  )) %>%
  mutate(star_cat = factor(star_cat, levels = c("stars_2.5_or_below", "stars

# Run regression for each year
models_list <- map(2010:2015, ~ lm(market_share ~ star_cat, data = filter(re

stargazer(models_list,
          type = "text",
          column.labels = as.character(2010:2015),
          dep.var.labels = "Market Share",
          covariate.labels = c("3 Stars", "3.5 Stars", "4 Stars", "4.5+ Star
          omit.stat = c("f", "ser"),
          title = "OLS Regression of Market Share on Star Ratings (2010-2015
          align = TRUE)
```

```
OLS Regression of Market Share on Star Ratings (2010-2015)
==================================================================
                              Dependent variable:
             ----------------------------------------------------
                                   Market Share
               2010      2011      2012      2013     2014      2015
               (1)       (2)       (3)       (4)      (5)       (6)
------------------------------------------------------------------
3 Stars       0.007*    0.010*    0.0001   0.016***  0.006    -0.005
             (0.004)   (0.005)   (0.005)   (0.005)  (0.011)  (0.007)

3.5 Stars    0.013***   0.009     -0.003   0.011**   0.006    0.002
             (0.004)   (0.005)   (0.005)   (0.005)  (0.011)  (0.007)

4 Stars      0.014***   0.006     -0.008    0.005    0.003    -0.006
             (0.004)   (0.006)   (0.005)   (0.005)  (0.011)  (0.007)

4.5+ Stars   0.019***  0.016***   -0.005    0.008   -0.002    -0.005
             (0.005)   (0.006)   (0.005)   (0.005)  (0.011)  (0.007)

Constant     0.013***  0.019***  0.030***  0.015***  0.019*   0.026***
             (0.003)   (0.005)   (0.004)   (0.004)  (0.010)  (0.007)

------------------------------------------------------------------
Observations  1,563     1,511     1,660     1,863    1,767    1,708
R2            0.015     0.005     0.003     0.008    0.005    0.004
Adjusted R2   0.013     0.002    0.0002     0.006    0.002    0.002
==================================================================
Note:                                  *p<0.1; **p<0.05; ***p<0.01
```

Question 5

In [19]:

```
rd_table_data <- ma_2010 %>%
  mutate(
    raw_score = rowMeans(
      across(any_of(rating.vars.2010) & where(is.numeric)),
      na.rm = TRUE
    )
  ) %>%
```

```
  filter(!is.na(raw_score)) %>%
  mutate(
    rounded_up = case_when(
      raw_score >= 2.75 & raw_score < 3.00 & Star_Rating == 3.0 ~ "3.0 Stars
      raw_score >= 3.25 & raw_score < 3.50 & Star_Rating == 3.5 ~ "3.5 Stars
      raw_score >= 3.75 & raw_score < 4.00 & Star_Rating == 4.0 ~ "4.0 Stars
      raw_score >= 4.25 & raw_score < 4.50 & Star_Rating == 4.5 ~ "4.5 Stars
      raw_score >= 4.75 & raw_score < 5.00 & Star_Rating == 5.0 ~ "5.0 Stars
      TRUE ~ NA_character_
    )
  )

rounding_summary <- rd_table_data %>%
  filter(!is.na(rounded_up)) %>%
  count(rounded_up, name = "Number of Plans Rounded Up") %>%
  rename(`Star Rating` = rounded_up)

rounding_summary
```

A tibble: 5 × 2

Star Rating	Number of Plans Rounded Up
<chr>	<int>
3.0 Stars	3808
3.5 Stars	4524
4.0 Stars	4077
4.5 Stars	773
5.0 Stars	30

Question 6

In [36]:

```
# Calculate raw scores
rd_table_data <- ma_2010 %>%
  mutate(
    raw_score = rowMeans(
      select(., any_of(rating.vars.2010)) %>% select(where(is.numeric)),
      na.rm = TRUE
    )
  ) %>%
  filter(!is.na(raw_score))

# Aggregate to plan-year level
rd_enrollment_data <- rd_table_data %>%
  group_by(contractid, planid) %>%
  summarize(
    raw_score = mean(raw_score, na.rm = TRUE),
    total_enrollment = sum(avg_enrollment, na.rm = TRUE),
    .groups = "drop"
  )
```

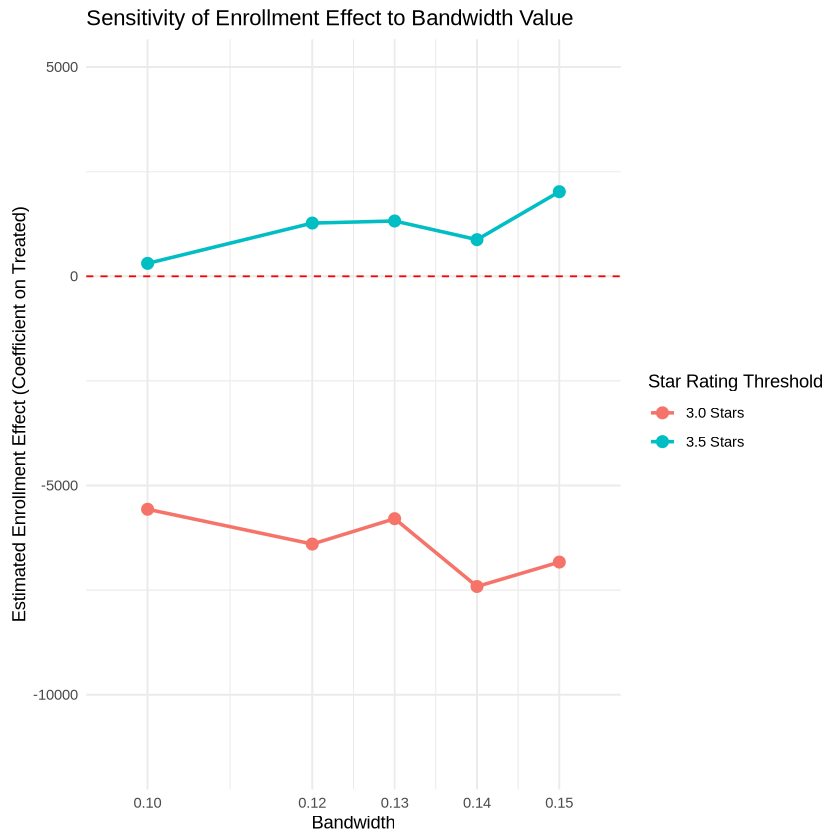
```
# RD estimation function (bandwidth = 0.125)
estimate_rd <- function(data, threshold, bw = 0.125) {
  data %>%
    mutate(dist = raw_score - threshold) %>%
    filter(abs(dist) <= bw) %>%
    mutate(treated = ifelse(dist >= 0, 1, 0)) %>%
    lm(total_enrollment ~ treated + dist, data = .)
}

# Run regressions for the rounding up thresholds
model_30 <- estimate_rd(rd_enrollment_data, 2.75)
model_35 <- estimate_rd(rd_enrollment_data, 3.25)

# Summary table using stargazer
stargazer(model_30, model_35,
          type = "text",
          column.labels = c("3.0 vs 2.5", "3.5 vs 3.0"),
          dep.var.labels = "Total Enrollments",
          covariate.labels = c("Treatment (Higher Star Rating)", "Running Va
          title = "RD Estimates using MA 2010 Data",
          omit.stat = c("f", "ser"))
```

```
RD Estimates using MA 2010 Data
============================================================
                                   Dependent variable:
                               ----------------------------
                                   Total Enrollments
                                 3.0 vs 2.5    3.5 vs 3.0
                                    (1)           (2)
------------------------------------------------------------
Treatment (Higher Star Rating)   -2,094.538     -269.304
                                (3,193.770)    (2,046.203)

Running Variable (Distance)      1,357.114     -5,820.447
                                (19,511.910)  (11,640.150)

Constant                        5,449.075***  3,799.438***
                                (1,790.259)    (1,148.518)

------------------------------------------------------------
Observations                        219            300
R2                                 0.008          0.008
Adjusted R2                       -0.002          0.001
============================================================
Note:                           *p<0.1; **p<0.05; ***p<0.01
```

Question 7

In [35]:
```
# Define parameters and bandwidths for analysis
thresholds <- c("3.0 Stars" = 2.75, "3.5 Stars" = 3.25)
bandwidths <- c(0.1, 0.12, 0.13, 0.14, 0.15)

# Create a function to run model and extract estimate
run_sensitivity <- function(thresh_val, thresh_label, bw) {
```

```
  model_data <- rd_enrollment_data %>%
    mutate(dist = raw_score - thresh_val) %>%
    filter(abs(dist) <= bw) %>%
    mutate(treated = ifelse(dist >= 0, 1, 0))

  fit <- lm(total_enrollment ~ treated + dist, data = model_data)

  # Data frame with estimate and standard error
  broom::tidy(fit, conf.int = TRUE) %>%
    filter(term == "treated") %>%
    mutate(bandwidth = bw, threshold = thresh_label)
}

# Run the model across all combos
sensitivity_results <- map2_dfr(
  rep(thresholds, each = length(bandwidths)),
  rep(names(thresholds), each = length(bandwidths)),
  ~run_sensitivity(.x, .y, bandwidths[match(.y, names(thresholds)) + (match(
)

# Correction for the map: cleaner version
sensitivity_results <- expand.grid(thresh_name = names(thresholds), bw = bar
  rowwise() %>%
  do(run_sensitivity(thresholds[[.$thresh_name]], .$thresh_name, .$bw))


# Create graph
ggplot(sensitivity_results, aes(x = bandwidth, y = estimate, color = threshc
    geom_line(size = 1) +
    geom_point(size = 3) +
    geom_errorbar(aes(ymin = conf.low, ymax = conf.high), linewidth = 0.005)
    scale_x_continuous(breaks = bandwidths) +
  theme_minimal() +
  labs(
    title = "Sensitivity of Enrollment Effect to Bandwidth Value",
    x = "Bandwidth",
    y = "Estimated Enrollment Effect (Coefficient on Treated)",
    color = "Star Rating Threshold"
  ) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red")
```

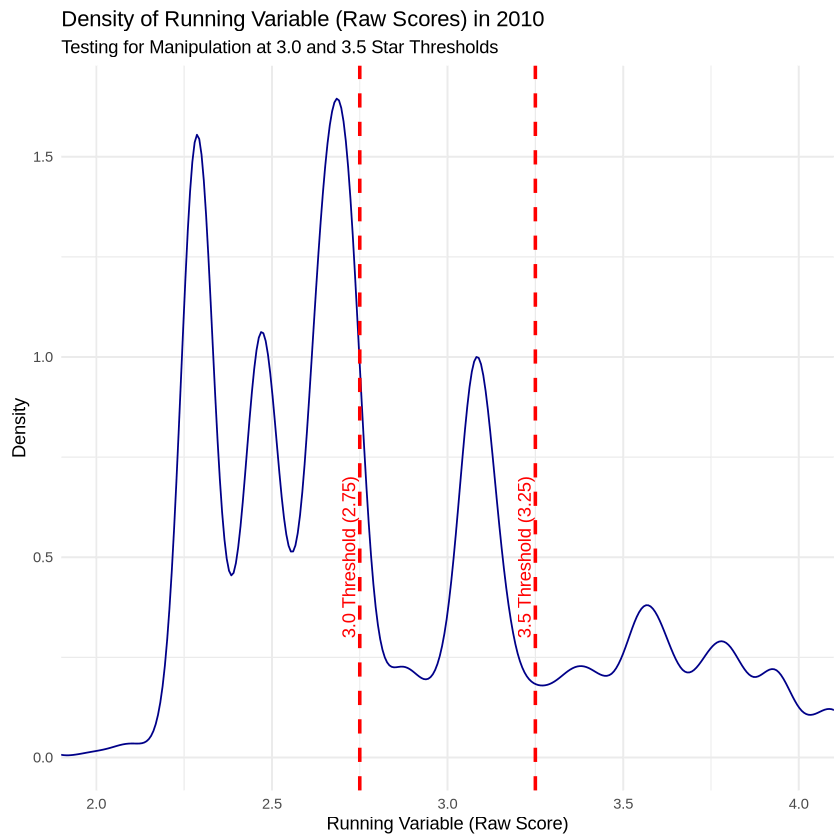


This bandwidth sensitivity analysis shows that RD estimates are robust to the choice of bandwidth. Regardless of window size (0.1 to 0.15), the estimated effect of a star rating increase remains statistically insignificant.

Question 8

In [30]:
```
# Define the relevant thresholds (rounding up for 3.0 and 3.5 stars)
thresholds <- c(2.75, 3.25)

# Create the combined density plot
ggplot(rd_table_data, aes(x = raw_score)) +
  geom_density( alpha = 0.3, color = "darkblue") +

  # Add threshold lines
  geom_vline(xintercept = thresholds,
             linetype = "dashed",
             color = "red",
             linewidth = 1) +

  # Add labels
  annotate("text", x = 2.70, y = 0.5, label = "3.0 Threshold (2.75)",
           angle = 90, color = "red", vjust = 1) +
  annotate("text", x = 3.20, y = 0.5, label = "3.5 Threshold (3.25)",
           angle = 90, color = "red", vjust = 1) +

  theme_minimal() +
  labs(
```

```
    title = "Density of Running Variable (Raw Scores) in 2010",
    subtitle = "Testing for Manipulation at 3.0 and 3.5 Star Thresholds",
    x = "Running Variable (Raw Score)",
    y = "Density"
  ) +

  coord_cartesian(xlim = c(2.0, 4.0))
```



The density plot shows a continuous distribution across the 2.75 and 3.25 thresholds (at which scores are rounded up to 3 and 3.5, respectively). There is no bunching or a discontinuous jump in the density of plans immediately following the rounding cutoffs. This supports the earlier findings that contracts did not strategically manipulate quality metrics to cross the thresholds in 2010.

Question 9

In [33]:
```
balance_data <- ma_2010 %>%
  mutate(
    raw_score = rowMeans(select(., any_of(rating.vars.2010)) %>% select(wher
    is_hmo = ifelse(grepl("HMO", plan_type, ignore.case = TRUE), 1, 0),
    has_partd = ifelse(partd == "Yes", 1, 0)
  ) %>%
  filter(!is.na(raw_score))

# RD Balance Test at 3.0 stars (threshold 2.75)
hmo_30 <- rdrobust(y = balance_data$is_hmo, x = balance_data$raw_score, c =
partd_30 <- rdrobust(y = balance_data$has_partd, x = balance_data$raw_score,
```

```
# RD Balance Test at 3.5 stars (threshold 3.25)
hmo_35 <- rdrobust(y = balance_data$is_hmo, x = balance_data$raw_score, c =
partd_35 <- rdrobust(y = balance_data$has_partd, x = balance_data$raw_score,
```

In [34]:
```
# Summary table of p-values for 3.0 and 3.5 thresholds
results_table <- data.frame(
  Characteristic = c("HMO Status", "Part D Status"),
  `P-Value at 3.0 Stars (2.75)` = c(hmo_30$pv[1], partd_30$pv[1]),
  `P-Value at 3.5 Stars (3.25)` = c(hmo_35$pv[1], partd_35$pv[1]),
  check.names = FALSE
)

results_table
```

A data.frame: 2 × 3

Characteristic	P-Value at 3.0 Stars (2.75)	P-Value at 3.5 Stars (3.25)
<chr>	<dbl>	<dbl>
HMO Status	0.000000e+00	0.000000e+00
Part D Status	1.506508e-252	9.266828e-16

Question 10

Looking at the results of analyzing 2010 MA data in questions 5-9, we can conclude that increasing star rating had no statistically significant effect on plan enrollment.

Question 6: We see that while we estimate a negative point estimate for both thresholds, neither is statistically significant. This suggests that in 2010, the enrollment effects at the 3.0 and 3.5 thresholds were statistically insignificant.

Question 7: Results remained relatively stable across bandwidths 0.1-0.15, proving the findings are robust.

Question 9: Both HMO status and Part D status show statistically significant discontinuities ($p < 0.01$) at the 2.75 and 3.25 thresholds. This indicates that plans above the rounding cutoffs were systematically different from those just below. Therefore, any observed effect on enrollment cannot be attributed solely to the rating, as these underlying plan characteristics likely confound the estimate.